

Feng Lin

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Education

Carnegie Mellon University

Expected: Dec 2026

School of Computer Science - Master of Science in Machine Learning

GPA: 4.00/4.00

University of North Carolina at Chapel Hill ([Thesis](#) – Highest Honors)

Aug 2021 - May 2025

Bachelor of Science: Economics & Statistics and Analytics (Double Majors); Cognitive Science (minor) GPA: 3.99/4.00

- **Areas:** Machine Learning, Multimodal Learning, Information Retrieval, Recommendation Systems, NLP, Computer Vision
- **Languages:** Python (PyTorch, TensorFlow, HuggingFace, Scikit-learn), MATLAB, SQL, R, Stata, GAMs
- **Tools:** Git, Shell, AWS, Weights & Biases, LaTeX

Working Experience

Shenwan Hongyuan Group Co., Ltd | Data Science Intern | Beijing

May - Aug 2024

- Built a quantitative risk-scoring model using **LSTM** and **ARIMA**, analyzing 3,000+ corporate financial statements to flag **120+ anomalies** from predicted trends across revenue, cash flow, and leverage metrics
- Designed a **multi-factor bond evaluation system** (ROA, interest coverage, quick ratio, etc.) that improved credit signal precision by 18% and automated data integration with an ETL pipeline for 50k+ entries to analyze issuers' financial status

Project Experience

LLM-Augmented Retrieval for Product Recommendation (Amazon Reviews 2023)

Jan 2026 – Present

- Built a **retrieval-based recommendation pipeline** using **Hypothetical Document Embeddings (HyDE)** to improve query representation for cold-start recommendation
- Generated **multimodal item embeddings** using **CLIP** and constructed a **FAISS-based vector index** for large-scale nearest-neighbor retrieval
- Generated synthetic LLM-generated user queries from user interaction history and reviews to improve retrieval quality, evaluated recommendation quality using Recall@K and NDCG@K

Single-Hop Question Answering with Classical and Transformer-based Retrieval Models

Aug 2025 - Dec 2025

- Built an **end-to-end single-hop QA pipeline**, benchmarking sparse retrievers (TF-IDF, BM25), sentence encoders, and BERT-family models (**BERT**, **ALBERT**, **RoBERTa**, etc.) under a unified evaluation framework
- Fine-tuned multiple transformer-based QA models on SQuAD 2.0 using **PyTorch**, analyzing pre- vs. post-fine-tuning gains in EM/F1, sentence-level accuracy and convergence speed through **wandb**
- Conducted systematic ablation analysis quantifying the impact of fine-tuning, **achieving up to 2.7x accuracy improvement in sentence retrieval (32.5% → 88.2%)** while keeping end-to-end inference time nearly constant

LLM-Based Hierarchical Video Understanding for Anomaly Detection

June 2025 - Aug 2025

- Developed a **hierarchical video understanding pipeline** using the VideoTree framework and GPT-4 reasoning to detect anomalies in long-form surveillance videos (UCF-Crime)
- Demonstrated improved temporal reasoning and efficiency for large-scale video understanding tasks. Improved **AUC from 0.742 to 0.853** while **reducing caption usage by 38%** and **end-to-end latency by 17%**

Saliency Maps for Semantic Segmentation in Dashcam Images

Aug 2024 - Dec 2024

- Integrated saliency maps from **CNN**-, **GAN**-, and **Transformer-based** models into a **U-Net segmentation** pipeline
- Improved car-class pixel accuracy by 13.1% and reduced pedestrian detection variance by 25%

American Bar Association Legal Service Supply Shortage

Mar 2024

- Built a two-stage risk-ranking pipeline integrating time-series prediction and specialty clustering to identify underserved regions with 82% recall, using models such as **LSTM**, **SVM**, and **KNN**
- Engineered a causality-informed feature that improved prioritization accuracy by 27% for downstream decision making

Publication

Lin, F., Dai, Y., Liu, Y. (2026). *On Probability Estimation for Unbalanced Classification*. Journal of Statistical Theory and Practice, Springer Nature. (Accepted)

- Introduces a posterior correction method to address probability calibration distortion caused by resampling and weighting

Lin, F., Wang, A. Z., Rahman, M. D., Szafer, D. A., Quadri, G. J. (2024). *Striking the Right Balance: Systematic Assessment of Evaluation Method Distribution Across Contribution Types*. BELIV Workshop, IEEE VIS. ([Paper](#))